RUCHITHA B M

Computer Science & Engineering | Full-Stack Developer | ML Enthusiast
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SUMMARY

Computer Science & Engineering student at Malnad College of Engineering, Hassan (expected 2027) with a strong problem-solving mindset and hands-on experience building full-stack MERN applications, an AI-based NLP chatbot, an AI-powered candidate-ranking system and a college-funded assistive-technology project. Familiar with DevOps and cloud tooling (Docker, Jenkins, AWS) and data visualization with Power BI. Active in hackathons and technical competitions.

TECHNICAL SKILLS

Frontend: React.js, HTML, CSS, JavaScript
Backend & APIs: Node.js, Express.js
Languages: Python, Java, JavaScript, Embedded C
Databases: MongoDB (Mongoose)
AI, ML & Data: Machine Learning, NLP, Power BI
Cloud & DevOps: AWS, Docker, Jenkins, GitHub
Soft Skills: Leadership & Teamwork, Technical Presentation, Time Management

EXPERIENCE

DevOps Intern, Crystallize Technologies, Hassan **Jan 2024 – Apr 2024**

- Containerized sample applications with Docker and deployed them on Kubernetes, writing Pod, Deployment and Service configurations and debugging issues with networking and external access.
- Built and tested CI/CD pipelines using Jenkins, Git and GitHub during structured training and a confidential internal company project, automating the build, test and deployment process.
- Explored container networking, load balancing and service exposure in lab exercises to understand how deployed applications stay reliable and reachable.
- Worked both independently and as part of a team to meet internship goals, applying DevOps concepts to practical problems.

PROJECTS

Smart Blind Stick with AI Voice Navigation — Funded Assistive Device for the Visually Impaired (Currently in Progress)
Embedded C | Arduino | Ultrasonic Sensor | GPS | GSM | IoT

- Team project shortlisted under NAIN 2.0 and awarded Rs. 3,80,000 in funding by Malnad College of Engineering; currently working on designing a low-cost smart stick to help visually impaired users move safely and independently.
- Designed a system with ultrasonic obstacle detection, voice and vibration alerts, GPS navigation, GSM-based SOS alerts with live location, water detection and solar charging.
- Surveyed 80 research papers (1997–2025) on smart blind sticks and identified gaps in indoor navigation, AI system cost, battery life and detection range.

MCE Cooperative Society Management System — Full-Stack Web Application
JavaScript | React.js | Node.js | Express.js | MongoDB

- Developed a cooperative society management system for Malnad College of Engineering with a responsive React front end and a Node.js/Express.js backend on MongoDB.
- Built inventory management for books, college records, accessories and society items, along with member management, transaction tracking and reporting modules.

AI Chatbot — MCE Cooperative Society Assistant — NLP-Based Query Assistant
Python | NLP | Machine Learning

- Built an AI-powered chatbot integrated with the MCE Cooperative Society Management System, letting members and staff query society info, inventory and records in natural language.
- Implemented intent recognition and context-aware responses to give instant answers and reduce manual lookup time.

Redrob Intelligent Ranker — AI-Powered Candidate Ranking System
Artificial Intelligence | HR Tech | Candidate Screening

- Designed an AI-powered candidate-ranking system that evaluates and prioritizes candidates against job requirements through a structured, two-phase workflow.
- Separated AI reasoning from application and data execution, using controlled application services to process requests and return structured ranking results.

Hostel Management System — Complaint Tracking Platform
JavaScript | React.js | Node.js | Express.js | MongoDB (Mongoose)

- Built a MERN-stack platform where students raise and track hostel complaints covering food quality, maintenance requests and facility problems.
- Developed an admin dashboard to view, assign and resolve issues, with user authentication, real-time status updates and notifications.

EDUCATION

Bachelor of Engineering (B.E.) in Computer Science & Engineering **Expected 2027**
Malnad College of Engineering, Hassan CGPA: 8.20
Diploma — SMT. L.V. Polytechnic, Hassan CGPA: 9.5
SSLC — Adichuchanagiri High School, K. Hirihally Score: 93%

CERTIFICATIONS

DevOps Certificate | Cloud Computing Certificate | Machine Learning Certificate | NPTEL Certificate

ACHIEVEMENTS

- Rs. 3,80,000 project funding from Malnad College of Engineering (NAIN 2.0); active in hackathons and technical competitions.